PROMPT ENGINEERING RESEARCH AND INTEGRATION INTERNSHIP

WEEK 4 — FINAL TEAM SUBMISSION

RIT 1409 PE TEAM 5

Included deliverables

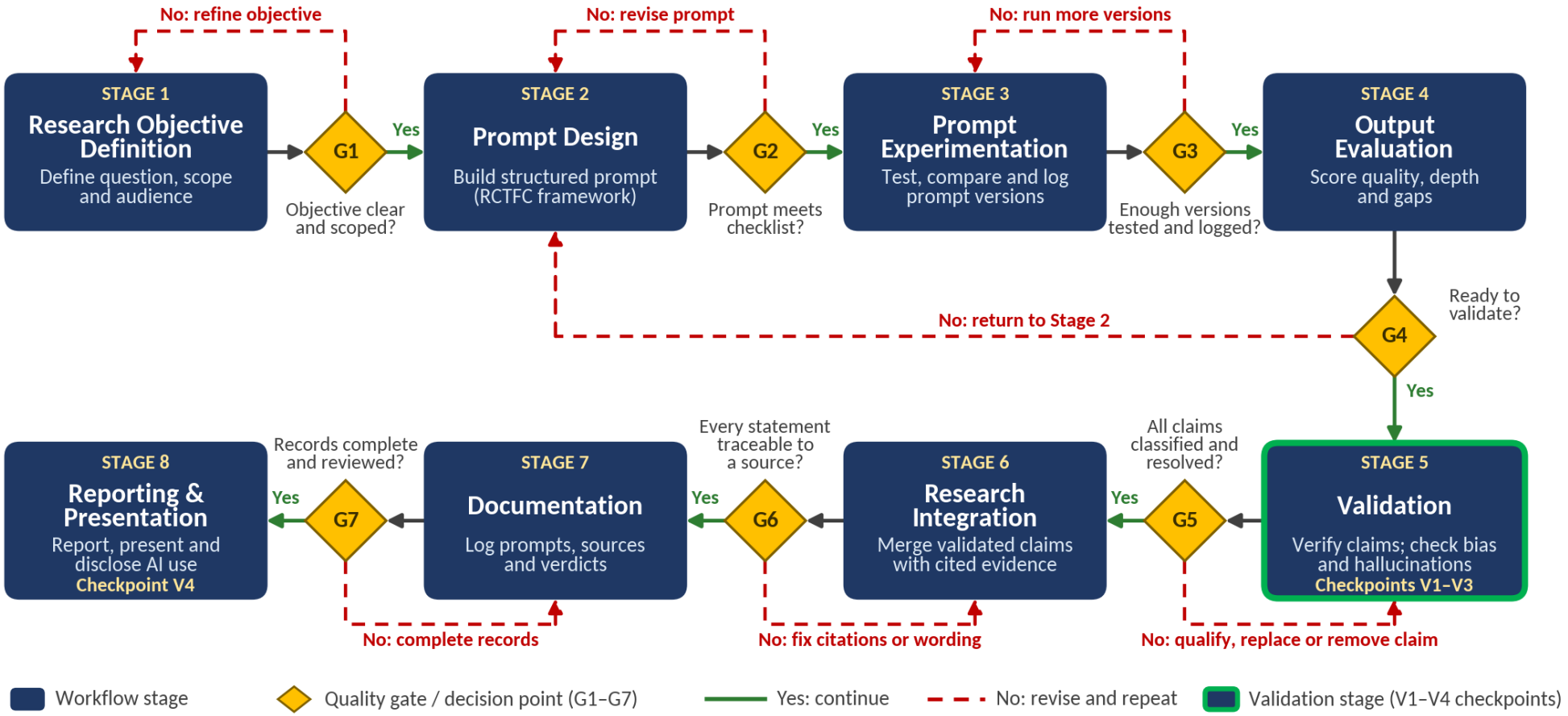
Deliverable	Included content
1. Workflow Blueprint	Eight-stage research workflow, quality gates, validation checkpoints, best practices and recommen
2. Prompt Repository	Ten prompts from the team's Week 1–2 work, with categories, objectives, frameworks, use cases a
Supporting guide	Prompt optimization, common mistakes, validation lessons and responsible AI practices.
3. Final Presentation	Presentation deck converted to PDF for convenient review; editable PPTX is supplied separately.

Prepared from the team's Week 1–4 materials. The separate PPTX is the editable presentation file. Any required video recordings and online evaluation forms must still be completed and submitted by the team.

AI-ASSISTED RESEARCH WORKFLOW BLUEPRINT

Week 4 — Deliverable 1

1. Workflow Diagram



2. Stage-by-Stage Explanation

Stage	What happens	Quality check	Output
1. Research Objective Definition	State the research topic and question. Define the audience, purpose and scope (region, time period) and what a good answer must contain.	Objective is specific, answerable and scoped. Audience and region are named (gap seen in Week 2 Prompt 1).	Written research objective with scope notes.
2. Prompt Design	Turn the objective into a structured prompt using a framework (RCTFC: Role, Context, Task, Format, Constraints). Ask for named sources and flagged uncertainty, and forbid invented citations.	Prompt checklist: all framework elements present, constraints stated, no vague terms.	Prompt version 1, saved word for word.
3. Prompt Experimentation	Run several versions on the same task (basic, structured, refined). Change one element at a time and record each version and its output.	Comparison criteria are set before reading the outputs; every version is logged.	Prompt version log and candidate outputs.
4. Output Evaluation	Score every output against the same criteria: structure, relevance, completeness, specificity, clarity, and visible gaps (audience, region, citations, definitions). Select the best version.	Same criteria applied to all versions; strengths and weaknesses written down.	Evaluation notes and selected output.
5. Validation	Extract 8–10 key factual claims (prioritise statistics, causal statements and absolute wording). Verify each against credible independent sources, then classify as Verified, Partially Verified or Unsupported/Unverified. Review for hallucinations, citation issues, bias and limitations.	Checkpoints V1–V3 passed (see Section 3). Wording, period, baseline and definition checked before any figure is called wrong.	Validation matrix and review notes.
6. Research Integration	Combine only validated or qualified statements with their supporting evidence. Rewrite Partially Verified claims to match the evidence, replace or remove Unsupported ones, and add citations.	Every statement is traceable to a checked source and is no stronger than that source.	Integrated, cited draft.
7. Documentation	Record the prompts and versions, outputs, evaluation notes, validation matrix, source list, decisions made and an AI-use/methodology statement.	Another person could repeat the process from the records; all links open.	Documentation pack / repository entries.
8. Reporting & Presentation	Produce the final report or deck with clear structure, visuals, limitations and recommendations. State how AI was used. Peer review before submission.	Checkpoint V4 (final check) passed; team review done.	Final report or presentation, correctly named and submitted.

3. Decision Points and Validation Checkpoints

Decision points (quality gates)

Gate	Decision question	If YES	If NO
G1	Is the objective specific, scoped and tied to an audience?	Go to Stage 2.	Refine the objective and scope.
G2	Does the prompt meet the design checklist (RCTFC, constraints, source request)?	Go to Stage 3.	Revise the prompt.
G3	Have enough versions been tested and logged for a fair comparison?	Go to Stage 4.	Run and log more versions.
G4	Does the best output meet the evaluation criteria and is it ready to validate?	Go to Stage 5.	Return to Stage 2 and redesign the prompt.
G5	Are all key claims classified, and have Partially Verified and Unsupported claims been resolved?	Go to Stage 6.	Qualify, replace or remove the claim, then re-verify. Re-prompt if needed.
G6	Is every statement traceable to a checked source and no stronger than the evidence?	Go to Stage 7.	Fix citations or wording and re-check against the validation matrix.
G7	Are the records complete, reproducible and peer reviewed?	Go to Stage 8.	Complete the records and repeat the review.

Validation checkpoints

Checkpoint	Stage	What is checked	Pass condition
V1 Claim verification	Stage 5	Each key claim is checked against an authoritative source. Check the exact wording, and for numbers check the period, baseline and definition. Do not mark Verified just because the general idea is true.	Every claim has a source, a verdict and a short explanation.
V2 Hallucination, bias and citation review	Stage 5	Look for invented sources or statistics, mixed-up causes, overstated or absolute wording, missing citations, one-sided framing and missing context.	Issues are listed and each is classed as poorly cited, overstated or unverifiable.
V3 Source and reference check	Stage 5–6	A second person confirms that each source exists, is relevant, supports the exact claim and has a working link.	No weak, missing or non-supporting sources remain.
V4 Final pre-submission check	Stage 8	Claims → sources → verdicts → bias analysis → references → formatting → names → filename.	Checklist fully ticked before submission.

What to do with each verdict

Verdict	Action in Stage 6
Verified	Use as written, with a citation.
Partially Verified	Keep only what the source supports. Correct the wording or add context (for example, “extinction risk rises with warming” instead of “species face extinction”).
Unsupported/Unverified	Do not use. Replace with a verified claim or remove it, and re-prompt if the gap matters.

Source standard: prefer IPCC, WHO, NOAA, NASA, UN agencies, government agencies and peer-reviewed research. Avoid blogs, Wikipedia, forums and personal websites.

4. Example: Applying the Workflow to the Climate-Change Task

Stage	What happened in our climate-change research
1. Objective	Explain the major impacts of climate change on humans and the environment.
2–3. Prompt design and experimentation	Prompt 1 was a basic single-line prompt with no role, format or constraints. More structured versions of the same task were then tested and compared.
4. Output evaluation	Prompt 1 was well structured and easy to scan but had no defined audience, no source citations and no regional specification.
5. Validation	Ten claims from Prompt 1 were checked against IPCC, NOAA and other institutional sources: 3 Verified, 7 Partially Verified, 0 Unsupported. For example, the answer linked ocean acidification to mass coral bleaching, but NOAA ties bleaching to heat stress, so the claim was qualified.
6. Integration	Validated and qualified wording, with citations, was carried into the research report instead of the original AI sentences.
7. Documentation	The validation matrix, verdicts, sources and the hallucination, citation and bias review were recorded.
8. Reporting	Findings were combined into the team report and final presentation, then checked before submission.

5. Best Practices and Recommendations

Prompting

- Define the objective, audience and scope (region, time period) before writing the prompt.
- Use a structured framework (RCTFC) and state constraints clearly. Ask the AI to name its sources and flag uncertainty, then check those sources yourself.
- Change one element at a time when experimenting, log every version, and evaluate all versions with the same criteria.

Validation

- Check statistics, causal statements and absolute words first (for example “unprecedented”, “all”, “will”).
- Check the period, baseline and definition before calling a figure wrong; sources often differ for these reasons.
- Verify the exact wording, not just the general idea, and have a second person check that each link supports the claim.

Responsible AI

- Treat AI output as a draft to be checked, never as evidence by itself.
- State how AI was used in every deliverable, and never invent or copy a citation without opening it.
- Do not enter personal or confidential information into prompts.
- Watch for bias and framing (loaded terms, one-sided emphasis, missing context) and state limitations honestly.

Documentation and recommendations

- Keep a prompt log, validation matrix and source list so the work can be reproduced.
- Reuse this blueprint as a checklist for future projects, and re-check figures that are updated between source editions.

References

IPCC (2021). *Climate Change 2021: The Physical Science Basis (AR6 Working Group I)*. https://www.ipcc.ch/report/ar6/wg1/

IPCC (2022). *Climate Change 2022: Impacts, Adaptation and Vulnerability (AR6 Working Group II)*. https://www.ipcc.ch/report/ar6/wg2/

NOAA National Ocean Service (n.d.). *What is Ocean Acidification?*. https://oceanservice.noaa.gov/facts/acidification.html

Prompt Collection

This document collects 10 optimized prompts developed by the team in Weeks 1 and 2. They are organized by category. For each prompt it gives the objective, category, framework, recommended use case and key optimization insights, followed by the prompt text as the team recorded it. Category coverage, best practices, common mistakes and lessons learned come after the prompts. Page numbers refer to the team's submitted PDFs.

Overview

No.	Prompt	Category	Source
1	Literature Review, Refined RCTFC (AI and Graduate Employability)	Research	Week 1, Prompt Design Logbook
2	Template: Academic Literature Synthesis	Research	Week 1, Prompt Framework Template Library
3	Climate Change Impacts (Detailed/Controlled RCTFC)	Research	Week 2, Prompt Experimentation Report
4	Pricing Analysis / Segmented Refinement (15% Price Increase)	Analysis	Week 1, Prompt Design Logbook
5	Template: Market Elasticity & Pricing Strategy	Analysis	Week 1, Prompt Framework Template Library
6	Competitive Analysis RCTFC Applied (Hypothetical SaaS Exercise)	Analysis	Week 1, Prompt Design Logbook
7	Template: Competitive Positioning & Differentiation	Analysis	Week 1, Prompt Framework Template Library
8	Template: Business Impact & Strategic Trend Analysis	Analysis	Week 1, Prompt Framework Template Library
9	Evidence Briefing, Refined Content Synthesis (Urban Heat Mitigation)	Reporting	Week 1, Prompt Iteration Reflection Report
10	Template: Multi-Source Evidence Synthesis	Reporting	Week 1, Prompt Framework Template Library

Category Coverage

The prompts are checked against the categories named in the Week 4 brief: Research, Analysis, Validation, Reporting and Evaluation. Only prompts that appear in the team's Week 1 to 3 files are counted.

Category	Prompts	Status
Research	3	Covered (Prompts 1 to 3).
Analysis	5	Covered (Prompts 4 to 8).
Reporting	2	Covered (Prompts 9 and 10).

Category	Prompts	Status
Validation	0	Not covered. No validation prompt is recorded in the team's files. Week 3 validation was done by checking claims against credible sources, using the team's validation format (Week 3 Work Plan, p.2) and a documented procedure (Week 3 report, p.8). These are methods, not prompts, so they are not counted. The Week 3 AI-use statement says Claude helped locate sources and draft validation text but does not record the prompts used (Week 3, p.9).
Evaluation	0	Not covered. No evaluation prompt is recorded in the team's files. Week 2 scored the outputs with a seven-criterion framework (Week 2, §5.1, p.9) and Week 1 used a self-built checklist (Week 1, p.13). These are scoring methods, not prompts, so they are not counted.

Category: Research

Prompt 1: Literature Review, Refined RCTFC (AI and Graduate Employability)

Source	Week 1, Prompt Design Logbook, Literature Review, Iteration 2 (Ayansiji Roseline Nike)
Objective	Review recent literature on the relationship between artificial intelligence and graduate employability, with particular attention to changing workplace skills, AI literacy, graduate preparedness, career development, opportunities, challenges, and gaps in existing research.
Category	Research
Framework	RCTFC (Role, Context, Task, Format, Constraints)
Recommended use case	Literature reviews on a defined topic and date range where study-level detail (author, year, methodology, findings), areas of agreement or disagreement, and research gaps are needed. Cited studies still need to be checked to confirm they exist and say what the output claims.
Key optimization insights	• Changes from the first version, as recorded in the logbook: a defined publication period (2020 to 2026) instead of "recent"; a request for study-level detail (author, year, focus, methodology, findings); a fixed comparison table; a request for agreement, disagreement and research gaps; and constraints on source quality and against invented studies or citations (Week 1, pp.5 to 6). • Several changes were made at once, so the improvement cannot be credited to any single one of them. The logbook records the result of one run.

Prompt used

Role: Academic research assistant with experience in conducting literature reviews and synthesizing scholarly research

Context: Research examines impact of AI on graduate employability, focusing on literature published between 2020 to 2026. Specific topics: changing workplace skills, AI literacy, graduate preparedness, career development, opportunities, challenges, areas of agreement/disagreement, research gaps

Task: Identify and synthesize relevant academic literature. Identify major themes emerging from literature. Compare findings across studies and identify areas where researchers agree or differ.

Format: Structured table with Author/Year, Research Focus, Methodology, Key Findings, and Relevance to Graduate Employability, followed by thematic synthesis and research gaps section

Constraints: Prioritize peer-reviewed and traceable academic sources published 2020 to 2026. Do not invent studies, authors, findings, publication dates, or citations. Distinguish empirical studies from conceptual papers and institutional reports.

Prompt 2: Template: Academic Literature Synthesis

Source	Week 1, Prompt Framework Template Library, Template 2
Objective	Conduct systematic reviews of academic literature with study-level detail, identify research gaps, and synthesize findings.
Category	Research
Framework	RCTFC (Role, Context, Task, Format, Constraints)
Recommended use case	Reusable template for any literature review. Replace the research topic and date range, and add geographic or disciplinary boundaries if relevant. The structured table makes it easy to compare studies and spot patterns.
Key optimization insights	• Generalizes the refined Literature Review prompt (Prompt 1): keeps the study table, the defined scope and the constraint against invented studies, and turns topic, date range and themes into placeholders (Week 1, pp.9 to 10). • Adds a constraint not in the applied prompt: do not cite sources that were not provided or verified, and flag known limitations of the studies reviewed (Week 1, pp.9 to 10). • No separate test run of the template is recorded in the Week 1 file.

Prompt used

Role: Academic research assistant with experience in conducting literature reviews, evaluating study quality, and synthesizing scholarly research across multiple sources.

Context: Research topic: [specific research question]. Literature scope: [define date range, geographic focus, discipline/field]. Key themes to explore: [list main topics of interest].

Task: Identify and synthesize relevant academic literature on [topic]. Extract and compare findings across studies. Identify major themes, areas of agreement and disagreement among researchers. Distinguish empirical research from conceptual papers. Identify research gaps and unanswered questions.

Format: (1) Structured table with columns: Author/Year, Research Focus, Methodology, Key Findings, Relevance to [topic]; (2) Thematic synthesis section identifying major findings and areas of agreement/disagreement; (3) Research gaps section identifying unanswered questions or limitations in current evidence.

Constraints: Prioritize peer-reviewed academic sources within the defined date range. Do not invent studies, authors, findings, or publication dates. Do not cite sources not specifically provided or verified. Distinguish study type: empirical research vs. conceptual paper vs. review vs. institutional report. Flag any known limitations of the studies reviewed.

Prompt 3: Climate Change Impacts (Detailed/Controlled RCTFC)

Source	Week 2, Prompt Experimentation Report, Prompt 3 (Chiamaka Chinaza)
Objective	Explain the major impacts of climate change on humans and the environment (environmental, social, economic and health), in a fixed section structure, using credible evidence, without invented statistics or sources, and with facts distinguished from uncertainty.
Category	Research
Framework	RCTFC, detailed/controlled variant (as labelled in the Week 2 report)
Recommended use case	Evidence-based overviews of a scientific or policy topic that need a fixed structure (Introduction, Key Findings, Impacts, Limitations, Conclusion). Statistics in the output still need to be checked against their sources and reference periods.

Key optimization insights	• Differs from the basic and structured climate prompts in several elements at once: an exact section list, an anti-fabrication instruction, and a request to separate facts from uncertainty (Week 2, §4.5, p.8). The higher Week 2 evaluation scores therefore cannot be credited to one element. • None of the three Week 2 prompts required inline attribution, so a reader cannot tell which source or reference period supports each figure (Week 2, §6.3, p.13). • In Week 3 the detailed figures were largely supportable but needed baselines, reference periods and careful causal wording. The team treated anti-hallucination instructions as risk reduction, not a guarantee (Week 3, pp.5 and 7). • The Week 2 results come from one set of outputs, scored by one team member (Week 2, §5, p.9).

Prompt used

"Act as an environmental research analyst, examine how climate change affects humans and the environment. Analyze the environmental, social, economic, and health impacts of climate change. Organize the answer into Introduction, Key Findings, Impacts, Limitations, and Conclusion. Use credible evidence/sources, do not invent statistics or sources, distinguish facts from uncertainty and avoid unsupported conclusions.

Category: Analysis

Prompt 4: Pricing Analysis / Segmented Refinement (15% Price Increase)

Source	Week 1, Prompt Design Logbook, Market Research, Iteration 3 (Asare Amma Miriam)
Objective	Estimate the revenue impact of a 15% price increase on a top-selling item separately for each customer segment, state whether the increase is advisable, and propose a lower-risk alternative if it is not.
Category	Analysis
Framework	RCTFC (Role, Context, Task, Format, Constraints)
Recommended use case	Pricing-change decisions where customer groups differ in price sensitivity (for example loyal customers versus price-comparison shoppers). The figures in the prompt ($40, 2,000 units, 60/40 segment split, competitor prices) are illustrative inputs to replace with real business data.
Key optimization insights	• Compared with the first RCTFC pricing version, this version splits customers into two segments, asks for revenue by segment instead of one blended demand curve, and asks for a named alternative strategy if the increase is risky (Week 1, p.4). • In the logbook's run, the blended break-even looked survivable, but the segmented analysis showed the price-sensitive segment alone pushing revenue down (Week 1, p.4, cross-iteration observation). • The elasticities and the customer split are assumptions, and the inputs are illustrative. The result comes from one run, so it shows how the prompt behaves, not a market finding.

Prompt used

Role: Pricing strategy consultant specializing in e-commerce revenue optimization

Context: Online store sells top-selling item currently priced at $40, with average monthly sales volume of 2,000 units. Two distinct customer segments: (1) repeat/loyal customers (~60% of sales, low price sensitivity) and (2) one-time/price-comparison shoppers (~40% of sales, high price sensitivity, price-checking against 2 competitors at $38 to $42). Goal: raise price by 15% (to $46)

Task: Estimate price increase impact on each customer segment separately. Recalculate revenue impact by segment rather than as one blended demand curve. State whether increase is advisable; if not, suggest alternative pricing move (smaller increase, segmented pricing, bundling) that could achieve similar revenue goal with less risk

Format: (1) Segment-by-segment demand impact table; (2) Blended revenue outcome under multiple elasticity scenarios; (3) Recommendation with named alternative strategy if straight 15% increase is risky

Constraints: Keep all assumptions explicit and labeled. Limit total response to 300 words excluding tables.

Prompt 5: Template: Market Elasticity & Pricing Strategy

Source	Week 1, Prompt Framework Template Library, Template 1
Objective	Analyze price sensitivity and demand elasticity for specific products or pricing changes.
Category	Analysis
Framework	RCTFC (Role, Context, Task, Format, Constraints)
Recommended use case	Reusable template for any proposed price change. Fill in the bracketed placeholders with the product details, current price, sales volume and target segment, and add competitor pricing if available.
Key optimization insights	• Generalizes the pricing prompts (Prompt 4): industry, price, volume, segment, competitors and brand loyalty become placeholders (Week 1, pp.8 to 9). • Adds a constraint not in the applied prompts: if data is unavailable, state what additional information would improve the analysis. Format item (4) invites segmented or alternative pricing, which matches the direction of the Iteration 3 change (Week 1, pp.4 and 8 to 9). • No separate test run of the template is recorded in the Week 1 file.

Prompt used

Role: Pricing strategy consultant specializing in [industry] revenue optimization. Your expertise includes demand analysis, competitive pricing, and revenue modeling.

Context: [Current product/service description]: Currently priced at [current price], with [monthly/annual sales volume] units sold. Target market segment: [segment description luxury, mid-market, commoditized]. Competitive landscape: [describe competitor pricing and positioning]. Brand position: [strong/moderate/weak loyalty].

Task: Analyze the impact of [proposed price change] on demand and revenue. Explain relevant pricing concepts (e.g., price elasticity of demand). Estimate plausible elasticity ranges for this product type. Calculate break-even points and revenue outcomes under different demand scenarios.

Format: (1) Explanation of relevant pricing concept [max 200 words]; (2) Scenario table showing current revenue vs. projected revenue at different demand-change levels; (3) Final recommendation with reasoning; (4) If applicable, suggest segmented or alternative pricing strategies.

Constraints: Keep all assumptions explicit and labeled as assumptions, not facts. Avoid generic disclaimers; focus on quantitative reasoning and the math. If data is unavailable, state what additional information would improve the analysis.

Prompt 6: Competitive Analysis RCTFC Applied (Hypothetical SaaS Exercise)

Source	Week 1, Prompt Design Logbook, Competitive Analysis, Iteration 2 (team compilation; hypothetical exercise)
Objective	Conduct a competitive analysis that identifies key competitors, compares product features and pricing strategies, evaluates market positioning, and reveals differentiation gaps.
Category	Analysis
Framework	RCTFC (Role, Context, Task, Format, Constraints)

Recommended use case	Competitive positioning analysis of a product against named rivals. This prompt was written for a hypothetical exercise, so competitor details produced from it are illustrative assumptions, not verified findings.
Key optimization insights	• Changes recorded in the logbook: an explicit strategist role; specific context (target segment, named competitors, geography, time horizon); a structured format with comparison tables; and constraints requiring unsourced claims to be framed as assumptions (Week 1, p.8). • The constraints ask the model to separate (a) publicly available facts, (b) inferences from product websites or reviews and (c) illustrative assumptions (Week 1, p.7). • This is a hypothetical exercise: its outputs are illustrative and were not checked against real competitor data (Week 1, p.6, Competitive Analysis disclaimer).

Prompt used

Role: Competitive strategist analyzing market positioning for a mid-market SaaS productivity platform

Context: [HYPOTHETICAL] Our platform targets mid-market teams (50–500 employees) seeking project management and team collaboration. Direct competitors: Asana, Monday.com, and ClickUp. Geographic focus: North America and Western Europe. Time horizon: current market conditions (2024–2026).

Task: Compare our product positioning against Asana, Monday.com, and ClickUp across: (1) core features (task management, team communication, reporting, integrations); (2) pricing model and per-seat cost at different team sizes; (3) market positioning (who they target, key marketing messages); (4) known weaknesses (based on customer reviews or known gaps); (5) our differentiation gaps (what they do better; what we do uniquely).

Format: (1) Feature comparison table showing presence/strength of key capabilities; (2) Pricing comparison for 50, 250, and 500-person teams; (3) Positioning matrix identifying each competitor's target segment and core value proposition; (4) Summary of competitive gaps and recommendations for differentiation.

Constraints: This is a hypothetical exercise. Use only publicly available information about these products where verifiable. Where specific competitor data is unavailable, frame claims as illustrative assumptions (e.g., "Assumed feature X" or "Based on publicly available descriptions") rather than stating them as verified facts. Do not invent unsourced pricing or capabilities. Clearly distinguish between: (a) publicly available facts, (b) inferences from product websites/reviews, and (c) illustrative assumptions for this exercise.

Prompt 7: Template: Competitive Positioning & Differentiation

Source	Week 1, Prompt Framework Template Library, Template 3
Objective	Analyze competitor strategies, identify market positioning, and reveal differentiation opportunities.
Category	Analysis
Framework	RCTFC (Role, Context, Task, Format, Constraints)
Recommended use case	Reusable template for comparing your product or service with its direct competitors. Identify your top 3–5 competitors and fill in the context with your product description and target market. Pair it with recent customer research or market reports, and keep verified facts separate from assumptions.
Key optimization insights	• Generalizes the competitive prompt (Prompt 6) with placeholders for market, competitors and positioning (Week 1, pp.10 to 11). • Keeps the rule that claims not based on documented sources are framed as assumptions or inferences, and adds identifying the source type and information gaps (Week 1, p.10). • The template's instructions suggest combining it with recent customer research or market reports (Week 1, p.11). No separate test run is recorded.

Prompt used

Role: Competitive strategist analyzing market positioning and identifying strategic differentiation opportunities for [your company/product].

Context: Your market: [describe target customer segment, geography, time horizon]. Direct competitors: [name 3–5 key competitors]. Your current positioning: [brief description of what your product/service does and who it serves].

Task: Compare your positioning against each competitor across: (1) core features/capabilities; (2) pricing and cost structure; (3) target market segment and marketing messages; (4) known strengths and weaknesses (from reviews, case studies, customer feedback); (5) your differentiation gaps.

Format: (1) Feature comparison table showing presence/strength of key capabilities across all competitors; (2) Pricing comparison table showing cost at different scales; (3) Positioning matrix or summary identifying each competitor's target segment and core value proposition; (4) Competitive gaps analysis identifying your advantages and weaknesses relative to each competitor.

Constraints: Limit analysis to publicly available, verifiable information. Do not invent features, pricing, or capabilities. For any claim not based on documented sources, explicitly frame it as an assumption or inference (e.g., "Assumed based on public product descriptions" or "Inferred from customer reviews"). Clearly identify the source type: product websites, published reviews, customer feedback, or industry reports. Identify any gaps in available information.

Prompt 8: Template: Business Impact & Strategic Trend Analysis

Source	Week 1, Prompt Framework Template Library, Template 5
Objective	Analyze policy shifts or workplace/market trends, quantify business impacts, and develop strategic recommendations.
Category	Analysis
Framework	RCTFC (Role, Context, Task, Format, Constraints)
Recommended use case	Reusable template for assessing a specific policy or trend (for example remote work adoption, AI deployment, or skill gaps in hiring) and its effect on an organization. Gather 3–5 credible sources with quantified data beforehand so the figures and benchmarks can be cited.
Key optimization insights	• Constraints require verified, traceable empirical data from credible sources, separation of observed data from expert interpretation, and a 500-word limit (Week 1, pp.12 to 13). • This lines up with the Week 3 recommendation to tie quantitative claims to their sources (Week 3, p.6). • The template's instructions say to gather 3 to 5 credible sources with quantified data before using it (Week 1, p.13). No separate test run is recorded.

Prompt used

Role: Senior business analyst or strategic advisor analyzing organizational and market trends using verified empirical data and developing evidence-based recommendations.

Context: Trend or policy: [describe the specific workplace trend, policy shift, or market change]. Relevance: [why this matters to your organization]. Scope: [which business functions, regions, or employee populations are affected]. Time horizon: [how rapidly is this changing].

Task: Analyze the direct impacts of [trend/policy] on [your business areas] using verified empirical data. Identify key organizational implications. Quantify business outcomes where data exists (cost impacts, efficiency gains, retention effects, etc.). Develop strategic recommendations for responding to this trend.

Format: (1) Trend overview (1–2 paragraphs); (2) Organizational impacts (3–4 key impact areas with supporting data); (3) Strategic recommendations (2–3 concrete actions); (4) Metrics for measuring effectiveness.

Constraints: Use only verified, traceable empirical data from credible sources (research institutions, government agencies, reputable surveys). Do not invent statistics or sources. Clearly distinguish between observed data and expert interpretation. Keep response concise (500 words or fewer) and focused on business-relevant impacts.

Category: Reporting

Prompt 9: Evidence Briefing, Refined Content Synthesis (Urban Heat Mitigation)

Source	Week 1, Prompt Iteration Reflection Report, Refined Prompt (Hassan Ridwan)
Objective	Help a city planning team understand what the evidence on urban heat mitigation supports and what to be cautious about, by combining several sources into one briefing.
Category	Reporting
Framework	RCTFC (Role, Context, Task, Format, Constraints)
Recommended use case	Turning a set of sources into a short internal decision briefing, especially when the sources partly disagree or one covers a related but different topic.
Key optimization insights	• The original prompt ("Combine these sources into a briefing.") had no rules for disagreement or off-topic sources. In the report's test it used a pollution finding to support a heat claim (Week 1, p.13). • The refined prompt added a role, audience and purpose, rules for handling disagreement and uncertainty, a fixed briefing structure and a word limit. The report states the refined output flagged the off-topic source and kept the disagreement visible (Week 1, pp.13 to 14). • The author's checklist scored the original 4/18 and the refined version 18/18 (Week 1, pp.13 to 14). The author notes the limits: one topic, a self-built source packet and self-scoring, so this is not independent evidence (Week 1, p.14).

Prompt used

Role: Content writer and researcher who turns information from multiple sources into clear, useful content. Your job is to understand how information fits together, not just summarize article by article.

Context: Topic—urban heat mitigation. Audience—city planning team making infrastructure decisions. Purpose—help decision-makers understand what evidence supports and what to be cautious about. Includes: real research, one open expert disagreement, one source on related but different topic.

Task: Read sources as a group. Identify main ideas, overlap, disagreement, and what readers need. Build coherent synthesis around central idea. Connect complementary sources; preserve genuine disagreements.

Format: Internal decision briefing. Max ~650 words. Structure: Bottom line / What evidence supports / Where it doesn't line up / What cannot be concluded / Implications / Evidence notes.

Constraints: Use only provided sources. Don't invent information. Don't repeat just because multiple sources mention it. Don't hide disagreements. Only name source when claim depends on where it came from. Prioritize usefulness over completeness.

Prompt 10: Template: Multi-Source Evidence Synthesis

Source	Week 1, Prompt Framework Template Library, Template 4
Objective	Combine multiple sources of evidence into clear, decision-focused briefings while distinguishing relevant from irrelevant information and preserving genuine disagreements.
Category	Reporting
Framework	RCTFC (Role, Context, Task, Format, Constraints)
Recommended use case	Reusable template for briefing a reader from several sources. Provide 4–6 sources, label any known disagreements between them, and state the audience and the decision the briefing should support.

Key optimization insights	• Generalizes the urban-heat briefing prompt (Prompt 9): topic, audience, purpose, sources and known disagreements become placeholders (Week 1, pp.11 to 12). • Keeps the fixed briefing structure and the rules to use only provided sources and not hide disagreements (Week 1, pp.11 to 12). • The template's instructions say to provide 4 to 6 sources and label any known disagreements (Week 1, p.12). It rests on one tested topic (Prompt 9) and no separate test run of the template is recorded.

Prompt used

Role: Content writer and researcher who turns information from multiple sources into clear, useful, reader-focused content. Your job is to understand how information fits together and communicate the bigger picture naturally—the way a reporter pieces together a real story after interviewing multiple sources on the same event.

Context: Topic: [specific decision or research question]. Audience: [who will read this and what do they need to decide?]. Purpose: [what decision or action should this briefing support?]. Sources provided: [list sources A, B, C, etc. with brief descriptions]. Background: [note any known disagreements, controversies, or relevant context among sources].

Task: Read the sources as a group. Identify main ideas, useful context, overlap, meaningful disagreements, and what the reader actually needs. Build one coherent synthesis around the central idea, not the order sources were given. Connect complementary sources; preserve genuine disagreements rather than forcing agreement.

Format: Internal decision-making briefing. Structure: Bottom line / What the evidence supports / Where it doesn't line up / What cannot be concluded / Implications / Evidence notes. Max ~650 words.

Constraints: Use only the provided sources for claims. Don't invent information. Don't repeat an idea just because multiple sources mention it. Don't hide disagreements or present uncertain claims as settled. Only name a source directly when the claim depends on where it came from. Prioritize usefulness over completeness.

Best Practices

1. Name the exact output sections you want. A defined structure was associated with a more complete answer than a loose "use clear headings" instruction (Week 2, §6.4, p.13).

2. Define scope concretely. Give dates, audience, named competitors and customer segments instead of words like "recent" (Week 1, pp.4, 5 to 6, 7 and 13).

3. Tell the model not to invent statistics, sources or citations, and still verify. Anti-fabrication wording is risk reduction, not a guarantee (Week 2, §6.4, p.13; Week 3, p.7).

4. Ask for inline attribution. Put the source name and reference year or period next to each figure, not in a list at the end (Week 2, §6.3 to 6.4, p.13).

5. Pair category names with a depth requirement. For example, "at least two evidenced points per category" (Week 2, §6.4, p.13).

6. Label assumptions. Keep verified facts, inferences and illustrative assumptions separate (Week 1, pp.3 to 4, 7 to 8 and 10 to 11).

7. Keep disagreement and uncertainty visible. Ask the model to preserve genuine disagreements and flag sources that do not answer the question (Week 1, pp.13 to 14).

8. State baselines and reference periods. Separate observed facts, projections and interpretations (Week 3, p.6).

9. Verify the exact wording of each claim before using it. Responsibility for final accuracy stays with the human researcher (Week 3, pp.6 to 7 and 9).

10. Test more than once and have someone else evaluate. Feedback on earlier weeks asked for repeated testing and independent evaluation. The urban-heat test was self-scored (Week 1, p.14).

Common Mistakes to Avoid

1. Writing a vague one-line prompt. With no role, context, format or constraints, the pricing and competitive prompts produced generic output (Week 1, pp.2 to 3 and 7).

2. Leaving the time scope undefined. “Recent academic research” did not define a publication period (Week 1, p.5).

3. Not prohibiting invented studies or citations. The first literature-review prompt did not (Week 1, p.5).

4. Listing sources at the end only. Readers cannot tell which source supports which claim (Week 2, §6.3, p.13).

5. Naming categories without asking for depth. The structured climate prompt returned about one paragraph per category (Week 2, §6.2, p.12).

6. Using one blended average for groups that behave differently. The blended pricing view hid the risk in the price-sensitive segment (Week 1, p.4).

7. Letting a related but different source support a claim. A pollution finding was used to back a heat claim in the original briefing (Week 1, p.13).

8. Treating AI output as evidence. Verification reduces hallucination risk but does not remove it (Week 3, pp.5 and 9).

9. Using universal wording or mixing causes. For example, linking ocean acidification directly to mass coral bleaching (Week 3, p.6).

10. Calling a figure wrong without checking its baseline or reference period. Different valid figures can come from different dates or definitions (Week 3 Work Plan, p.3; Week 2, §5.3, p.10).

11. Crediting one prompt element when several changed at once, or drawing strong conclusions from a single set of outputs. This was raised in the feedback on the earlier weeks.

12. Presenting hypothetical competitor data as fact. The competitive analysis was an illustrative exercise (Week 1, p.6, Competitive Analysis disclaimer).

Lessons Learned

1. Combined changes give combined results. Because the prompt versions changed several elements together, the results show what each full version produced, not which single element did the work.

2. Accuracy and attribution are separate. A response can be factually correct and still hard to verify without its source and reference period (Week 2, §6.3, p.13 and §7, p.14).

3. Anti-hallucination instructions lower risk but do not remove it. Detailed outputs still needed claim-level verification (Week 3, pp.5 to 7).

4. General answers are easy to check but carry little evidence. The structured climate output was generally supported but comparatively general (Week 3, p.5).

5. Segmenting a problem can reveal risk a blended view hides. This was the main difference between the two pricing versions (Week 1, p.4).

6. Rules about evidence matter as much as writing quality. How the prompt treats disagreement, uncertainty and unequal evidence shaped the urban-heat briefing (Week 1, p.14).

7. Current results rest on small samples. They come from single runs, one or a few outputs per prompt, ten selected claims per prompt in Week 3, and in places self-scoring (Week 1, p.14; Week 3, p.7). Repeated testing and independent evaluation are the next step.

8. The human team stays responsible. AI output is material to check, and final accuracy remains with the researchers (Week 3, p.9).

Here are the key insights regarding prompt optimization, RCTFC lessons, best practices, common mistakes, validation lessons, and overall findings:

1. Prompt Optimization & Performance Across Iterations

- **Prompt 1 (Basic/Simple):** Produced useful broad insights, but suffered from the least controlled wording. **7 out of 10 claims were only "Partially Verified"** due to missing citations, overstatements, missing reference periods, and mixed causal mechanisms.
- **Prompt 2 (Structured RCTFC):** Produced cleaner, well-structured outputs that were 100% verified across all 10 claims. However, its main limitation was **low evidentiary specificity**—the claims were accurate but broad and low in specific detail.
- **Prompt 3 (Detailed/Controlled RCTFC):** Generated rich, quantitative, research-style data with specific metrics and baselines. While highly informative, it demonstrated that **high-detail prompts require strict claim-level verification**, as specific metrics introduced reference-period ambiguities.

2. RCTFC Prompting Framework Lessons

- **Role/Role-play:** Establishing a clear persona (e.g., climate research assistant) ensures the tone aligns with formal research standards.
- **Context:** Providing tight contextual boundaries keeps outputs focused, but broad contexts can lead to unintentional overgeneralization.
- **Task:** Explicit tasks prevent vagueness; structuring tasks into distinct categories yields higher verification rates.
- **Format:** Enforcing structured deliverables (e.g., 4-category breakdowns) improves readability and factual containment.
- **Constraints (Crucial Anti-Hallucination Mechanism):** Explicit anti-fabrication instructions, uncertainty flags, and requirements for sources/baselines significantly reduce hallucination risks —though they serve as **risk reduction, not absolute guarantees**.

3. Common AI Mistakes & Hallucination Risk Factors

- **Causal Conflation:** The AI frequently mixed distinct mechanisms (e.g., linking ocean acidification directly as a driver of coral bleaching, whereas thermal heat stress is the direct driver of bleaching).
- **Overstatement & Universalization:** Using absolute terms (e.g., claiming *all* tropical cyclones get stronger, or species unable to adapt face *guaranteed* extinction) rather than nuanced, evidence-matched language.

- **Reference-Period Ambiguity:** Presenting statistics without baselines (e.g., stating sea-level rise figures without specifying whether the period is 1880–present vs. 1901–2018) leads to apparent discrepancies.
- **Terminology Inaccuracy:** Using informal or legally imprecise terms (e.g., "climate refugees") as absolute categories rather than technically precise language (e.g., "climate-related displacement").
- **Missing Source Traceability:** Simple prompts tend to generate plausible quantitative figures without inline citation anchors, making verification difficult.

4. Best Practices for AI-Assisted Research

1. **Require explicit baselines & dates:** Force the prompt to specify start years, comparison periods, and reference standards for any statistic.
2. **Mandate mechanism separation:** Instruct the AI to delineate distinct physical processes (e.g., thermal stress vs. chemical acidification).
3. **Incorporate uncertainty qualifiers:** Require the AI to flag projections vs. observed facts and express conditional probability rather than absolute certainty.
4. **Enforce regional & contextual specificity:** Prevent uniform framing by requiring regional qualifiers (e.g., "in low-lying coastal zones").
5. **Treat AI outputs as material to validate, never as evidence:** AI-generated claims must always be treated as draft hypotheses requiring independent verification against primary institutional sources (e.g., IPCC, NOAA, WHO).

5. Research Validation Lessons

- **Claim-Level Granularity:** Validation cannot just check the "general idea"; wording precision, numbers, dates, baselines, and causality must be evaluated line by line.
- **"Partially Verified" Category is Essential:** Many claims are not outright fabrications but require qualification due to exaggerated wording or missing context.
- **Human-in-the-Loop Imperative:** Anti-hallucination prompting minimizes risk, but human domain-expert verification remains mandatory for factual accuracy and final submission integrity.

What We Learned from Improving Prompts

1. **Trade-off Between Specificity and Risk**: Simple/unstructured prompts yield high-level, unchecked claims with wording errors. Structured prompts (RCTFC) yield safe, broad claims but lack depth. Detailed, controlled prompts produce rich quantitative content, but increase the necessity for claim-level verification of reference periods and statistics.

2. **Prompts Function as Risk Reduction Tools**: Prompt engineering cannot replace validation; it serves as a pre-filtering mechanism to minimize framing errors and missing citations before human review.
3. **Structured Research Workflow**: Responsible AI use follows a strict lifecycle:

Generate structured draft → Extract factual claims → Verify against credible sources → Refine wording/dates/baselines → Retain human accountability

RIT 1409 | PE TEAM 5

AI Prompt Engineering

From prompt design to research validation

AI Prompt Engineering Research & Integration Internship

Final presentation • Weeks 1–4

THE PROGRESSION

Our four-week journey

01

Prompt design

Build reusable RCTFC prompts and refine a multi-source synthesis.

02

Prompt experiments

Run three prompts on the same climate-change question and score outputs.

03

Research validation

Check selected claims, examine framing and qualify wording.

04

Workflow design

Turn the lessons into an eight-stage process and reusable prompt collection.

The central question stayed the same: how can prompt design improve AI-assisted research, and what still needs human checking?

The RCTFC framework

R	**Role**	What expertise or perspective should the AI use?
C	**Context**	What background, audience, scope and assumptions matter?
T	**Task**	What exact work must be done?
F	**Format**	What structure should the response follow?
C	**Constraints**	What rules govern sources, uncertainty, length and invention?

A framework improves the prompt’s instructions. It does not verify the answer by itself.

Source: Week 1 Prompt Framework Template Library

Week 1: three research scenarios

Market research

A hypothetical online store considers a 15% price rise on a best-selling product. Prompt iterations add price, volume, elasticity assumptions and customer segments.

Literature review

A review of AI and graduate employability, scoped to 2020–2026, moving toward a study-by-study table of methods, findings and gaps.

Competitive analysis

A hypothetical SaaS platform is compared with Asana, Monday.com and ClickUp. The specific competitor claims were illustrative, not verified.

The team also created five reusable RCTFC templates: market research, literature review, competitive analysis, content synthesis, and policy/trend analysis.

Source: Week 1 Prompt Design Logbook

WEEK 1 APPLIED EXAMPLE

Pricing example: context changes the analysis

Iteration 1 · Vague

“An online store wants to raise prices by 15%. Will this work?”

Output: general explanation of price elasticity and a suggestion to test on a small scale.

Iteration 2 · Structured

Adds a $40 price, 2,000 monthly units, a $46 target price, a scenario table and explicit assumptions.

Break-even sales decline calculated at about 13.04%.

Iteration 3 · Segmented

Separates loyal customers from price-sensitive shoppers and tests segment-specific assumptions.

The model’s base case showed blended revenue falling 10.9%, so the recommendation changed.

Lesson: more context enabled a more specific analysis. The numerical outputs depend on the scenario’s assumptions; they are not observed store results.

Source: Week 1 Market Research Logbook; all business inputs below are scenario assumptions

Content synthesis: 4/18 to 18/18

4 / 18
Original prompt

18 / 18
Refined RCTFC prompt

What changed

The original prompt combined five urban-heat sources without clear rules. It used a related air-pollution finding as evidence for heat and glossed over disagreement about heat measurement.

What the refined prompt did

It flagged the irrelevant source, preserved the disagreement and stated the trade-off around trees, pedestrian heat stress and air ventilation.

Important limitation

The checklist was created and scored by the report's author, using one topic and a source packet assembled for the exercise.

The scores show the result of this exercise, not proof that the same improvement will occur for every topic.

Source: Prompt Iteration Reflection Report

Week 2: one question, three prompts

Shared task: “Explain the major impacts of climate change on humans and the environment.”

Prompt 1 · Basic

A one-line request with no role, defined output format or constraints. The answer was organized, but figures lacked source attribution.

Prompt 2 · Structured RCTFC

Named environmental, social, economic and health categories and asked for evidence-based information. Clear headings, but limited depth.

Prompt 3 · Detailed RCTFC

Specified Introduction, Key Findings, Impacts, Limitations and Conclusion, with instructions not to invent sources or statistics and to separate fact from uncertainty.

The task was held constant; the prompts differed in their level of structure and detail.

Source: Week 2 Prompt Experimentation Report

Output evaluation matrix

Criterion	Prompt 1	Prompt 2	Prompt 3
Accuracy	3	3	5
Clarity	4	4	4
Completeness	3	2	5
Relevance	4	4	5
Structure	3	3	5
Hallucination risk*	3	3	4
Bias risk*	4	4	4

*Risk scales run in the opposite direction: 5 = very low risk. Scores are the team's evaluation, not an independent benchmark.

Source: Output Evaluation Matrix , scores 1–5

How to read the scores

For hallucination and bias risk, a higher score means lower assessed risk.

Important qualification

Prompt 3's accuracy score reflected the Week 2 evaluation. Week 3 still found two of its ten selected claims needed qualification.

What the experiment taught us

What helped

Exact output sections improved completeness and structure.

Anti-fabrication instructions helped reduce risk.

Categories worked better when paired with a required level of depth.

What remained weak

No prompt consistently required source, year and reference period beside each claim.

Broad categories could produce shallow paragraphs.

A role alone showed no clear isolated effect.

The distinction

Evaluation compares the usefulness and quality of outputs.

Validation checks whether specific claims are supported by credible evidence.

Source: Week 2 Prompt Optimization Recommendations

RESEARCH VALIDATION METHOD

Week 3: how we checked the claims

1

Select

10 key claims from each prompt output

2

Check

Compare exact wording with credible independent sources

3

Classify

Verified, Partially Verified, or Unsupported/Unverified

4

Explain

Record source, verdict and why wording needed correction

What mattered most

Check the exact claim, especially statistics, dates, baselines, causal statements and absolute wording. A generally true idea does not automatically validate every phrasing of it.

Source: Week 3 Research Validation Matrix and Hallucination & Bias Report

Validation results: 30 selected claims

Prompt output	Verified	Partially verified	Unsupported / unverified	Total
Prompt 1 · Basic	3	7	0	10
Prompt 2 · Structured	10	0	0	10
Prompt 3 · Detailed	8	2	0	10
Total	21	9	0	30

Verified

The checked source supported the claim as written.

Partially verified

The main idea held, but wording, causality, scope or reference period needed qualification.

These results cover the 30 selected claims, not every sentence in the three AI outputs.

Source: Week 3 Research Validation Matrix; 10 selected claims per prompt

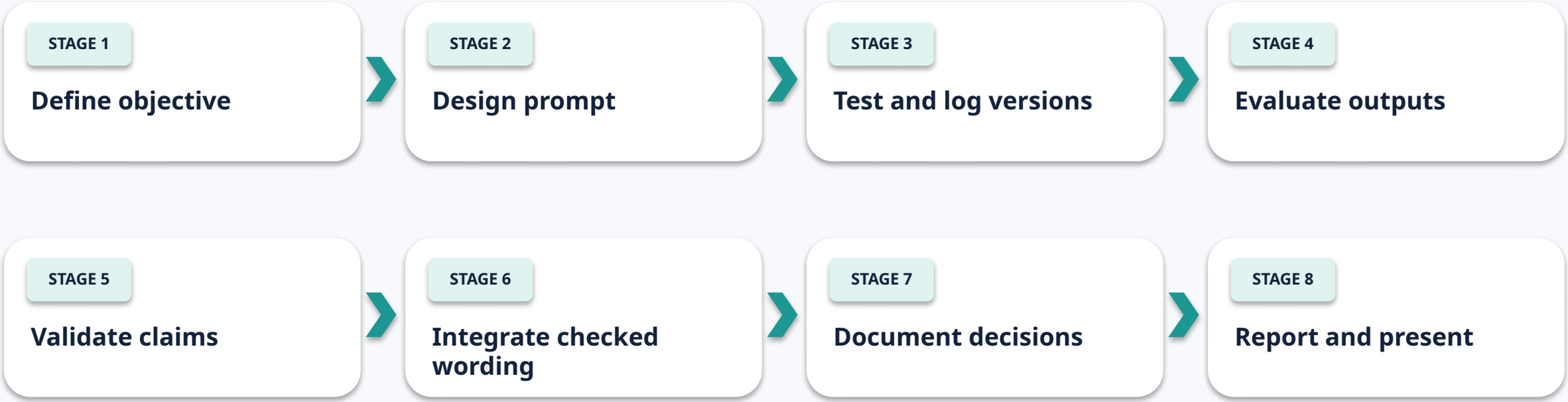
What needed closer attention?

Missing traceability	Prompt 1 included figures without inline sources.
Overstatement	Universal claims about cyclones or species survival exceeded the evidence as worded.
Reference periods	Climate figures can differ because dates, baselines or definitions differ.
Causal conflation	Ocean acidification and heat-stress-driven coral bleaching are distinct processes.
Terminology and framing	“Climate-related displacement” is more precise than treating “climate refugee” as a universal legal status.

No fabricated source, quotation, study or event was identified among the selected claims. This does not prove every unreviewed sentence was correct.

Source: Week 3 Hallucination & Bias Detection Report

Week 4: an eight-stage research workflow



Decision gates ask whether the objective, prompt, output, claims and records are ready to move forward. If not, revise and re-check.

The blueprint also defines four validation checkpoints, including a second-person source/link check and a final pre-submission review.

Source: Week 4 AI-Assisted Research Workflow Blueprint

The prompt collection

RESEARCH	3 prompts	AI and graduate employability literature review; academic literature synthesis template; climate-change impacts prompt.
ANALYSIS	5 prompts	Pricing and elasticity; market pricing template; hypothetical SaaS competitive analysis; competitive positioning template; business impact and trend analysis template.
REPORTING	2 prompts	Refined urban-heat evidence briefing; multi-source evidence synthesis template.

Total: 10 prompts. Validation and evaluation were methods used in the project, not separate prompt categories in the repository.

Source: Week 4 Prompt Collection / Repository

WHAT WE WOULD DO NEXT

Recommendations for future research tasks

Specify the output

Name the sections and how much evidence each should contain.

Make evidence traceable

Require a source, date and reference period beside each quantitative claim.

Test fairly

Change one prompt element at a time, log versions and use the same scoring criteria.

Verify the exact claim

Check numbers, causal wording, definitions and scope against the source.

Keep uncertainty visible

Separate observations, projections and interpretations; qualify claims by region where needed.

Retain human accountability

Treat AI output as a draft, document AI use and review the final claims.

Sources: Week 2 recommendations and Week 4 workflow blueprint

Limitations and responsible AI use

Limits of the exercise

- One main research topic: climate change
- Three prompt versions in the Week 2 comparison
- Ten selected claims checked per prompt
- No repeated runs of the same prompts
- No independent second scoring of the same outputs

AI use and accountability

AI was used in the Week 2 prompt experiments and Week 3 validation materials. The Week 3 report documents Claude in Roseline's process for locating sources and drafting validation text.

The team checked AI outputs and remained responsible for the final claims. AI output was treated as a draft, not evidence by itself.

These limits mean the findings are useful lessons from a structured exercise, not proof that one prompt design will always outperform another.

Sources: Week 3 AI-Assisted Research Report; Week 4 Workflow Blueprint

TRACEABILITY

Selected references and source documents

INTERNAL TEAM DOCUMENTS

RIT 1409 PE Team 5 (2026). Week 1 Prompt Design Logbook, Prompt Framework Template Library and Prompt Iteration Reflection Report.

RIT 1409 PE Team 5 (2026). Week 2 Prompt Experimentation Report and Output Evaluation Matrix.

RIT 1409 PE Team 5 (2026). Week 3 Research Validation Matrix, Hallucination & Bias Detection Report and AI-Assisted Research Report.

RIT 1409 PE Team 5 (2026). Week 4 AI-Assisted Research Workflow Blueprint and Prompt Collection.

EXTERNAL RESEARCH SOURCES

IPCC (2021). Climate Change 2021: The Physical Science Basis. https://www.ipcc.ch/report/ar6/wg1/

IPCC (2022). Climate Change 2022: Impacts, Adaptation and Vulnerability. https://www.ipcc.ch/report/ar6/wg2/

WHO. Climate Change and Health. https://www.who.int/news-room/fact-sheets/detail/climate-change-and-health

NOAA. What is Ocean Acidification? https://oceanservice.noaa.gov/facts/acidification.html

NOAA. How does climate change affect coral reefs? https://oceanservice.noaa.gov/facts/coralreef-climate.html

NASA Science. Ice Sheets. https://science.nasa.gov/earth/explore/earth-indicators/ice-sheets

IDMC. Global Report on Internal Displacement. https://www.internal-displacement.org/global-report/

Hong Kong Observatory. Climate Change. https://www.hko.gov.hk/en/climate_change/

Selected references listed in the submitted Week 1–4 documents

Thank You

RIT 1409 PE Team 5

AI Prompt Engineering Research & Integration Internship